

COS1501

UNIT 3

SETS

Symbols and terminology first • then complete guide-level theory
Every main topic includes a medium and a hard fully worked example

Colour key

TEAL definitions and laws	BLUE medium worked examples	PURPLE hard worked examples	RED / ORANGE traps and exam notes
------------------------------	-----------------------------------	--------------------------------	--------------------------------------

Part A — Learn the symbols before the theory

Set theory becomes difficult when symbols are read vaguely. Read each expression as a complete mathematical sentence. The terminology column gives the exact name and spoken reading.

EXAM NOTE - How to read a set statement

First identify the objects, then the relation or operation, then the domain. For example, $3 \in A$ says an object belongs to a set; $\{3\} \subseteq A$ says a set is contained in a set.

SYMBOL	TERMINOLOGY / READ AS	MEANING	BRIEF EXAMPLE
$\{ \}$	braces / curly brackets	Enclose the elements or defining rule of a set.	$\{1,2,3\}$
,	comma / separator	Separates distinct listed elements.	$\{a,b,\{c\}\}$ has three elements.
...	ellipsis / "and so on"	Continues an evident pattern; unsafe if ambiguous.	$\{2,4,6,\dots\}$
	such that	Separates a variable/domain from its admission rule.	$\{x \in \mathbb{Z} \mid x^2=4\}$
$\in$	is an element of	Relates one object to a set.	$2 \in \{1,2,3\}$
$\notin$	is not an element of	States failed membership.	$4 \notin \{1,2,3\}$
$=$	set equality	Both sets have exactly the same elements.	$\{1,2\} = \{2,1,1\}$
$\neq$	is not equal to	The sets differ by at least one element.	$\{1\} \neq \{\{1\}\}$

Symbol language — containment and operations

SYMBOL	TERMINOLOGY / READ AS	MEANING	BRIEF EXAMPLE
$\subseteq$	is a subset of	Every left-set element belongs to the right set.	$\{1,2\} \subseteq \{1,2,3\}$
$\subset$	is a proper subset of	Subset but not equal.	$\{1,2\} \subset \{1,2,3\}$
$\not\subseteq$	is not a subset of	A left-side element is absent from the right set.	$\{1,4\} \not\subseteq \{1,2,3\}$
$\cup$	union	Elements in A or B, including both.	$\{1,2\} \cup \{2,3\} = \{1,2,3\}$
$\cap$	intersection	Elements common to both sets.	$\{1,2\} \cap \{2,3\} = \{2\}$
$-$	difference / relative complement	Elements in A but not in B.	$\{1,2\} - \{2,3\} = \{1\}$
A'	complement of A	Elements of U that are not in A.	$U = \{1,2,3\}, A = \{1\}$: $A' = \{2,3\}$
$A+B$	symmetric difference	Elements in exactly one of A and B.	$\{1,2\} + \{2,3\} = \{1,3\}$

Symbol language — special sets and counting

SYMBOL	TERMINOLOGY / READ AS	MEANING	BRIEF EXAMPLE
$\emptyset$	empty set	The unique set with no elements; $\emptyset = \{ \}$.	$ \emptyset =0$
U	universal set	All objects currently under discussion.	Complements are taken inside U.
 A 	cardinality of A	Number of distinct elements in finite A.	$ \{a,b,b\} =2$
n(A)	cardinality of A	Alternative notation for A .	$n(\{0,1\})=2$
P(A)	power set of A	The set of all subsets of A.	$P(\{a\})=\{\emptyset,\{a\}\}$
2ⁿ	two to the power n	Number of subsets of an n-element set.	$ A =3 \Rightarrow P(A) =8$
iff	if and only if	Both implications hold; a reversible condition.	$A=B$ iff $A \subseteq B$ and $B \subseteq A$
$\Rightarrow$	implies	The left statement guarantees the right.	$x \in A \cap B \Rightarrow x \in A$

SYMBOL	TERMINOLOGY / READ AS	MEANING	BRIEF EXAMPLE
$\mathbb{Z}^+$	positive integers	1,2,3,...	$3 \in \mathbb{Z}^+$
$\mathbb{Z}_{\geq 0}$	non-negative integers	0,1,2,3,...	$0 \in \mathbb{Z}_{\geq 0}$
$\mathbb{Z}$	integers	..., -2, -1, 0, 1, 2, ...	$-5 \in \mathbb{Z}$
$\mathbb{Q}$	rational numbers	Ratios p/q, with integers p,q and $q \neq 0$.	$3/7 \in \mathbb{Q}$
$\mathbb{R}$	real numbers	All rational and irrational numbers.	$\sqrt{2} \in \mathbb{R}$

Part B — Terminology you must use correctly

Definitions are not loose descriptions. Each term fixes a logical condition that must be used in explanations and proofs.

TERM	PRECISE MEANING
set	A well-defined collection of distinct objects.
element / member	An individual object belonging to a set.
well-defined	Membership can be decided unambiguously for every candidate object.
list / roster notation	Lists elements inside braces.
set-builder notation	Uses $\{x \in D \mid P(x)\}$: variable, domain and property.
property description	The condition $P(x)$ deciding whether x enters.
universal set	The complete domain U relative to the discussion.
empty set	The set $\emptyset$ with no elements.
subset	$A \subseteq B$ means every A -element is a B -element.
proper subset	$A \subset B$ means $A \subseteq B$ and $A \neq B$.
disjoint sets	Sets with no common element: $A \cap B = \emptyset$.
cardinality	Number of distinct elements of a finite set.
power set	$P(A)$, the set of every subset of A .
inclusive or	A or B or both; used by union.
exclusive or	A or B , but not both; symmetric difference.

Unit roadmap and required level

Unit 3 prepares the language used later for relations, functions and proofs. You must translate accurately between words, symbols, roster notation and set-builder notation; distinguish membership from containment; and compute new sets without losing the universal-set context.

EXAM NOTE - Guide-level outcomes

Define every Unit 3 term; prove or disprove containment and equality; form union, intersection, difference, complement and symmetric difference; decide disjointness; calculate finite cardinalities; and construct power sets.

PART	FOCUS	COMPLETE COVERAGE
1	Describing sets	well-defined sets, roster form, ellipsis, set-builder form
2	Comparing sets	membership, equality, nesting, empty and universal sets
3	Containment	subsets, proper subsets, counterexamples
4	Set operations	union, intersection, difference, complement, symmetric difference
5	Size and sets of sets	disjointness, cardinality, inclusion-exclusion, power sets

- Can you decide whether an object is an element, a subset, both or neither?
- Can you justify equality with mutual containment?
- Can you track the universal set through complements and mixed operations?
- Can you distinguish a power set from the original set and count it correctly?

1. Sets, elements and well-defined collections

DEFINITION - Sets, elements and well-defined collections

A set is a well-defined collection of distinct objects. Those objects are its elements or members.

Objects may be numbers, letters, names, sets, or mixed objects. “Well-defined” means every proposed object has an objective yes/no membership answer.

A set records membership only. Order and repetition do not create new information.

NOTE - Admission test

Treat the description as an entrance requirement. If reasonable people could disagree without violating the wording, it is not well-defined.

MEDIUM WORKED EXAMPLE - Decide membership

Problem. Let A be the set of integer solutions of $x^2 - 5x + 6 = 0$. Find A and decide whether $2 \in A$.

Solution.

Factor: $(x-2)(x-3)=0$, so $A=\{2,3\}$. Therefore $2 \in A$.

HARD WORKED EXAMPLE - Repair an ambiguous collection

Problem. Explain why $S=\{x \in \mathbb{R} \mid x \text{ is close to } 0\}$ is not well-defined, then give a precise replacement containing -0.2 but not 0.8 .

Solution.

“Close” has no fixed threshold. One repair is $S=\{x \in \mathbb{R} \mid |x| < 1/2\}$. Then $|-0.2| < 0.5$, while $|0.8|$ is not below 0.5 .

2. List notation and the roster method

DEFINITION - List notation and the roster method

Roster notation lists every distinct element inside braces, separated by commas.

Use it when all elements can be listed or a finite pattern has an unmistakable stopping point. Simplify listed expressions before deciding whether they create new elements.

$$\{a, b, c\} = \{c, a, b, a\}$$

EXAM NOTE - Order and repetition

Sets are unordered and have no duplicate elements. Rearrangement or repetition does not change the set.

MEDIUM WORKED EXAMPLE - Translate to roster form

Problem. Write $\{x \in \mathbb{Z} \mid -3 \leq x < 2\}$ in roster notation.

Solution.

The allowed integers are $-3, -2, -1, 0, 1$. Hence $\{-3, -2, -1, 0, 1\}$.

HARD WORKED EXAMPLE - Simplify before counting

Problem. Write $A = \{0, 1 - 1, 2^2 - 4, \{0\}, \emptyset\}$ without repetitions and find $|A|$.

Solution.

The first three entries equal 0. The objects $\emptyset$ and $\{0\}$ are different. Thus $A = \{0, \{0\}, \emptyset\}$ and $|A| = 3$.

3. Ellipsis and ambiguous patterns

DEFINITION - Ellipsis and ambiguous patterns

An ellipsis (...) means continuation, but defines a set only when context uniquely fixes the rule.

The start $\{3,5,7,\dots\}$ could mean odd integers greater than 1, primes greater than 2, or many other sequences. Set-builder notation removes uncertainty.

A rigorous description provides both a domain and a property.

COMMON TRAP - Dots are not a rule

Never infer a unique infinite set from a few terms unless the surrounding statement fixes the pattern.

MEDIUM WORKED EXAMPLE - Replace dots precisely

Problem. Describe $\{4,8,12,16,\dots\}$ as the positive multiples of 4.

Solution.

$\{x \in \mathbb{Z}^+ \mid 4 \text{ divides } x\}$, equivalently $\{4k \mid k \in \mathbb{Z}^+\}$.

HARD WORKED EXAMPLE - Two valid continuations

Problem. Give two distinct well-defined infinite sets whose first four listed elements are 1,2,4,8.

Solution.

One is $\{2^n \mid n \in \mathbb{Z}_{\geq 0}\}$. Another is $\{1,2,4\} \cup \{x \in \mathbb{Z}^+ \mid x \geq 8\}$. Therefore the four entries alone do not determine the set.

4. Set-builder notation

DEFINITION - Set-builder notation

The form $\{x \in D \mid P(x)\}$ means all objects x from domain D such that property $P(x)$ is true.

The variable is a placeholder, so changing its name does not change the set. The domain prevents unintended objects from entering.

$$\{x \in \mathbb{Z} \mid -4 \leq x < 0\} = \{y \in \mathbb{Z} \mid -4 \leq y < 0\}$$

EXAM NOTE - State the domain

The equation $x^2=2$ over $\mathbb{Z}$ and over $\mathbb{R}$ defines different sets: $\emptyset$ versus $\{-\sqrt{2}, \sqrt{2}\}$.

MEDIUM WORKED EXAMPLE - Translate exactly

Problem. Write the even non-negative integers below 10 in set-builder form.

Solution.

$\{x \in \mathbb{Z}_{\geq 0} \mid x \text{ is even and } x < 10\}$, or $\{2k \mid k \in \mathbb{Z} \text{ and } 0 \leq k \leq 4\}$.

HARD WORKED EXAMPLE - Solve the admission rule

Problem. Convert $\{x \in \mathbb{R} \mid x^2 - 3x + 2 < 0\}$ to an inequality interval and justify it.

Solution.

Factor $(x-1)(x-2) < 0$. The factors have opposite signs strictly between the roots, so $\{x \in \mathbb{R} \mid 1 < x < 2\}$.

5. Membership: $\in$ and $\notin$

DEFINITION - Membership: $\in$ and $\notin$

$x \in A$ says object x is an element of A . $x \notin A$ is its negation.

Membership compares an object with a set. It does not assert that every member of one set lies in another; that is subset language.

Braces change the object: 1 , $\{1\}$ and $\{\{1\}\}$ are different.

COMMON TRAP - Type-check the statement

Ask whether the left side is one object and the right side is a set.

MEDIUM WORKED EXAMPLE - Read a nested set

Problem. Let $A = \{1, \{1\}, 2\}$. Decide whether $1 \in A$ and $\{1\} \in A$.

Solution.

Both are true: 1 is listed, and the set $\{1\}$ is separately listed.

HARD WORKED EXAMPLE - Classify four statements

Problem. Let $A = \{\emptyset, \{\emptyset\}\}$. Determine $\emptyset \in A$, $\emptyset \subseteq A$, $\{\emptyset\} \in A$ and $\{\{\emptyset\}\} \subseteq A$.

Solution.

All are true. $\emptyset$ and $\{\emptyset\}$ are listed. Also $\emptyset$ is a subset of every set. The only element of $\{\{\emptyset\}\}$ is $\{\emptyset\}$, which belongs to A .

6. Set equality

DEFINITION - Set equality

$A=B$ exactly when A and B have precisely the same elements; equivalently $A \subseteq B$ and $B \subseteq A$.

Order, repetition, variable names and description style do not matter. Only membership matters.

To disprove equality, one element in one set but not the other is sufficient.

EXAM NOTE - Proof method

For a non-obvious equality, prove both containments by translating membership conditions.

MEDIUM WORKED EXAMPLE - Check equality

Problem. Compare $A=\{-4,-3,-2,-1\}$ and $B=\{x \in \mathbb{Z} \mid -5 < x < 0\}$.

Solution.

The integers strictly between -5 and 0 are exactly the elements of A . Therefore $A=B$.

HARD WORKED EXAMPLE - Prove a polynomial description

Problem. Prove $\{x \in \mathbb{R} \mid 1 < x < 2\} = \{x \in \mathbb{R} \mid x^2 - 3x + 2 < 0\}$.

Solution.

Since $x^2 - 3x + 2 = (x-1)(x-2)$, the product is negative exactly between its roots. The membership conditions are equivalent, so the sets are equal.

7. Nested sets and levels of braces

DEFINITION - Nested sets and levels of braces

A set may itself be an element of another set. Each pair of braces creates one new set object.

Never remove braces by intuition. In $A = \{1, \{1\}\}$, the object 1 and the object $\{1\}$ are distinct.

Membership moves one level; subset compares elements at the current level.

NOTE - Count top-level entries

Count distinct top-level entries, not objects hidden inside inner braces.

MEDIUM WORKED EXAMPLE - Count correctly

Problem. Find the elements and cardinality of $A = \{\{1,2\}, \{2,1\}, 1\}$.

Solution.

Since $\{1,2\} = \{2,1\}$, the first two entries duplicate one set. Thus $A = \{\{1,2\}, 1\}$ and $|A| = 2$.

HARD WORKED EXAMPLE - Membership chain

Problem. Let $A = \{1, \{1\}, \{\{1\}\}\}$. Which of 1, $\{1\}$, $\{\{1\}\}$, $\{\{\{1\}\}\}$ belong to A?

Solution.

The first three are explicitly listed. The set $\{\{\{1\}\}\}$ is not, so only it fails membership.

8. The empty set

DEFINITION - The empty set

The empty set $\emptyset = \{ \}$ has no elements and is unique.

A set-builder rule defines $\emptyset$ when no object in its domain satisfies the property.

The empty set is a subset of every set because no element of $\emptyset$ can violate containment.

COMMON TRAP - Do not confuse

$\emptyset$ has 0 elements, while $\{\emptyset\}$ has 1 element—the empty set itself.

MEDIUM WORKED EXAMPLE - Identify an impossible rule

Problem. Evaluate $A = \{x \in \mathbb{Z} \mid x \text{ is odd and divisible by } 2\}$.

Solution.

No integer satisfies both conditions. Therefore $A = \emptyset$ and $|A| = 0$.

HARD WORKED EXAMPLE - Prove the universal subset fact

Problem. Prove $\emptyset \subseteq B$ for every set B .

Solution.

The statement means: if $x \in \emptyset$, then $x \in B$. There is no $x \in \emptyset$, hence no counterexample. So the implication holds for every B .

9. Universal sets

DEFINITION - Universal sets

A universal set U contains all objects currently under discussion.

The same named subset can have different complements under different universes. U is part of the data, not decoration.

Every set under discussion must be a subset of U .

EXAM NOTE - Context controls meaning

Always identify U before taking a complement.

MEDIUM WORKED EXAMPLE - Change the universe

Problem. Let $A = \{1, 3\}$. Find A' if $U = \{1, 2, 3, 4, 5\}$.

Solution.

$A' = U - A = \{2, 4, 5\}$.

HARD WORKED EXAMPLE - Two universes, two answers

Problem. Let A be the even integers. Find A' in $U = \mathbb{Z}$ and then in $U = \mathbb{Z} \geq 0$.

Solution.

In $\mathbb{Z}$, A' is all odd integers. In $\mathbb{Z} \geq 0$, A' is only the non-negative odd integers $\{1, 3, 5, \dots\}$.

10. Subsets

DEFINITION - Subsets

$A \subseteq B$ iff every element of A is also an element of B .

Equality is allowed: $A \subseteq A$. Also $\emptyset \subseteq B$ for every B .

To prove containment, start with arbitrary $x \in A$. To disprove it, produce $x \in A$ with $x \notin B$.

EXAM NOTE - One witness defeats containment

A single missing element proves $A \not\subseteq B$.

MEDIUM WORKED EXAMPLE - Test a subset

Problem. Is $\{2, 4\} \subseteq \{x \in \mathbb{Z} \mid x \text{ is even}\}$?

Solution.

Yes. Both left-side elements are even integers.

HARD WORKED EXAMPLE - Prove divisibility containment

Problem. Let $A = \{x \in \mathbb{Z} \mid 12 \text{ divides } x\}$ and $B = \{x \in \mathbb{Z} \mid 4 \text{ divides } x\}$. Prove $A \subseteq B$ and decide $B \subseteq A$.

Solution.

If $x = 12k$, then $x = 4(3k)$, so $A \subseteq B$. But $4 \in B$ and $4 \notin A$, hence $B \not\subseteq A$.

11. Proper subsets

DEFINITION - Proper subsets

$A \subset B$ means $A \subseteq B$ and $A \neq B$; at least one B -element is missing from A .

A finite proper subset has fewer elements than its containing finite set. Formally, use containment plus inequality.

No set is a proper subset of itself.

NOTE - Symbol convention

This guide uses $\subseteq$ for subset allowing equality and $\subset$ for proper subset.

MEDIUM WORKED EXAMPLE - Decide properness

Problem. Let $A = \{1, 3\}$ and $B = \{1, 2, 3\}$. Is $A \subset B$?

Solution.

Yes. Every A -element is in B , and $2 \in B - A$. Hence $A \subset B$.

HARD WORKED EXAMPLE - Infinite proper subset

Problem. Prove $\mathbb{Z}^+ \subset \mathbb{Z}$ although both sets are infinite.

Solution.

Every positive integer is an integer, so $\mathbb{Z}^+ \subseteq \mathbb{Z}$. But $0 \in \mathbb{Z}$ and $0 \notin \mathbb{Z}^+$, so the containment is strict.

12. Union and inclusive “or”

DEFINITION - Union and inclusive “or”

$A \cup B = \{x \mid x \in A \text{ or } x \in B\}$, where “or” includes the case “both”.

An element in both sets is included once. Union is commutative and associative, and $A \cup \emptyset = A$.

$$A \cup B = B \cup A; (A \cup B) \cup C = A \cup (B \cup C)$$

COMMON TRAP - Inclusive or

Do not remove common elements. Union includes the overlap.

MEDIUM WORKED EXAMPLE - Compute a union

Problem. Let $A = \{1, 2, 4\}$ and $B = \{2, 3, 4\}$. Find $A \cup B$.

Solution.

Collect all distinct elements in either set: $\{1, 2, 3, 4\}$.

HARD WORKED EXAMPLE - Union of descriptions

Problem. Let $A = \{x \in \mathbb{R} \mid x < 2\}$ and $B = \{x \in \mathbb{R} \mid x \geq -1\}$. Find $A \cup B$.

Solution.

Every real x satisfies at least one condition and the intervals overlap. Therefore $A \cup B = \mathbb{R}$.

13. Intersection

DEFINITION - Intersection

$$A \cap B = \{x \mid x \in A \text{ and } x \in B\}.$$

Intersection retains only common elements. It is commutative and associative; $A \cap U = A$ and $A \cap \emptyset = \emptyset$.

$$A \cap B = B \cap A; (A \cap B) \cap C = A \cap (B \cap C)$$

NOTE - And means both

An element failing either condition is excluded.

MEDIUM WORKED EXAMPLE - Compute an intersection

Problem. Let $A = \{1, 2, 4\}$ and $B = \{2, 3, 4\}$. Find $A \cap B$.

Solution.

The common elements are 2 and 4, so $A \cap B = \{2, 4\}$.

HARD WORKED EXAMPLE - Solve two conditions

Problem. Let $A = \{x \in \mathbb{Z} \mid 3 \text{ divides } x\}$ and $B = \{x \in \mathbb{Z} \mid 4 \text{ divides } x\}$. Describe $A \cap B$.

Solution.

The integers divisible by both 3 and 4 are exactly the multiples of $\text{lcm}(3, 4) = 12$. Thus $A \cap B = \{x \in \mathbb{Z} \mid 12 \text{ divides } x\}$.

14. Difference / relative complement

DEFINITION - Difference / relative complement

$$A-B = \{x \mid x \in A \text{ and } x \notin B\}.$$

Difference is directional: $A-B$ usually differs from $B-A$. It removes B -elements only from A .

$$A-B = A \cap B'$$

COMMON TRAP - Read left to right

Start with the left set, then delete every element also found in the right set.

MEDIUM WORKED EXAMPLE - Compute both directions

Problem. For $A = \{1, 2, 3, 4\}$ and $B = \{3, 4, 5\}$, find $A-B$ and $B-A$.

Solution.

$A-B = \{1, 2\}$, while $B-A = \{5\}$. Difference is not commutative.

HARD WORKED EXAMPLE - Difference of intervals

Problem. Let $A = \{x \in \mathbb{R} \mid -2 \leq x \leq 4\}$ and $B = \{x \in \mathbb{R} \mid 1 < x < 6\}$. Describe $A-B$.

Solution.

Keep A -points not in B . Values from -2 through 1 remain, including 1 . Thus $\{x \in \mathbb{R} \mid -2 \leq x \leq 1\}$.

15. Complement

DEFINITION - Complement

Relative to U , $A' = \{x \in U \mid x \notin A\} = U - A$.

Complement reverses membership inside a fixed universe. Double complement returns the original set.

$$(A')' = A; \quad U' = \emptyset; \quad \emptyset' = U$$

EXAM NOTE - De Morgan laws

$$(A \cup B)' = A' \cap B' \text{ and } (A \cap B)' = A' \cup B'.$$

MEDIUM WORKED EXAMPLE - Find a complement

Problem. Let $U = \{1, 2, 3, 4, 5, 6\}$ and $A = \{2, 4, 6\}$. Find A' .

Solution.

Delete A from U : $A' = \{1, 3, 5\}$.

HARD WORKED EXAMPLE - Prove De Morgan by elements

Problem. Prove $(A \cup B)' = A' \cap B'$.

Solution.

For $x \in U$: $x \in (A \cup B)'$ iff $x \notin A \cup B$ iff $x \notin A$ and $x \notin B$ iff $x \in A' \cap B'$. Thus the sets are equal.

16. Symmetric difference

DEFINITION - Symmetric difference

$A+B$ contains elements in exactly one of A and B , not both.

The guide writes $A+B$; many texts use $A\triangle B$. Both mean exclusive-or membership.

$$A+B = (A-B) \cup (B-A) = (A \cup B) - (A \cap B)$$

COMMON TRAP - Do not confuse with union

Union keeps the overlap; symmetric difference removes it.

MEDIUM WORKED EXAMPLE - Compute symmetric difference

Problem. Let $A = \{1, 2, 3\}$ and $B = \{3, 4, 5\}$. Find $A+B$.

Solution.

$A-B = \{1, 2\}$ and $B-A = \{4, 5\}$; therefore $A+B = \{1, 2, 4, 5\}$.

HARD WORKED EXAMPLE - Solve an unknown set

Problem. Let $U = \{1, 2, 3, 4, 5\}$, $A = \{1, 2, 3\}$, and $A+B = \{2, 4, 5\}$. Find B .

Solution.

Symmetric difference toggles membership. Thus $2 \notin B$; $4, 5 \in B$; and $1, 3 \in B$ because they are absent from $A+B$. Hence $B = \{1, 3, 4, 5\}$.

17. Disjoint sets

DEFINITION - Disjoint sets

A and B are disjoint iff $A \cap B = \emptyset$.

Disjointness means no shared elements; neither set needs to be empty.

If A and B are disjoint, then $A \cup B = A + B$ and finite cardinalities add.

NOTE - Quick test

Find the intersection. One common element destroys disjointness.

MEDIUM WORKED EXAMPLE - Check disjointness

Problem. Are $\{1,3,5\}$ and $\{2,4,6\}$ disjoint?

Solution.

Yes. Their intersection is $\emptyset$.

HARD WORKED EXAMPLE - Parameter condition

Problem. Let $A = \{x \in \mathbb{R} \mid x \leq a\}$ and $B = \{x \in \mathbb{R} \mid x > b\}$. For which relation are A and B disjoint?

Solution.

They overlap when some x satisfies $b < x \leq a$, possible exactly when $b < a$. Therefore they are disjoint exactly when $a \leq b$.

18. Cardinality

DEFINITION - Cardinality

For a finite set A , $|A|$ or $n(A)$ is its number of distinct elements.

Duplicates are ignored because they create no new element. Inclusion-exclusion corrects double-counting of overlap.

$$|A \cup B| = |A| + |B| - |A \cap B|$$

EXAM NOTE - Count distinct objects

Simplify equal entries and respect nested braces before counting.

MEDIUM WORKED EXAMPLE - Count a union

Problem. If $|A|=18$, $|B|=13$ and $|A \cap B|=5$, find $|A \cup B|$.

Solution.

$$|A \cup B| = 18 + 13 - 5 = 26.$$

HARD WORKED EXAMPLE - Recover the overlap

Problem. In a class of 60, 38 study Python, 31 study Java, and 8 study neither. How many study both?

Solution.

The union has 52. If the overlap is k , $52 = 38 + 31 - k$, so $k = 17$.

19. Power sets

DEFINITION - Power sets

$P(A)$ is the set of all subsets of A .

Its elements are sets. Both $\emptyset$ and A belong to $P(A)$. If $|A|=n$, then $|P(A)|=2^n$ because each element is included or excluded.

$X \in P(A)$ iff $X \subseteq A$

EXAM NOTE - Membership-subset bridge

To test $X \in P(A)$, test whether $X \subseteq A$.

MEDIUM WORKED EXAMPLE - Construct a power set

Problem. Find $P(\{a,b\})$.

Solution.

The subsets are $\emptyset, \{a\}, \{b\}, \{a,b\}$. Hence $P(\{a,b\}) = \{\emptyset, \{a\}, \{b\}, \{a,b\}\}$ and its cardinality is 4.

HARD WORKED EXAMPLE - Nested power set

Problem. Let $A = \{\emptyset, \{\emptyset\}\}$. List $P(A)$.

Solution.

A has two elements. Its subsets are $\emptyset, \{\emptyset\}, \{\{\emptyset\}\}$, and $\{\emptyset, \{\emptyset\}\}$. These four sets form $P(A)$.

20. Comparing power sets

DEFINITION - Comparing power sets

Power-set identities must be checked at the level of subset membership.

Exactly: $P(A \cap B) = P(A) \cap P(B)$. Generally, $P(A) \cup P(B)$ is only a subset of $P(A \cup B)$.

A mixed subset using an A-only and a B-only element belongs to $P(A \cup B)$ but to neither individual power set.

COMMON TRAP - Complement warning

If complementing $P(A)$, specify a universe of sets, often $P(U)$.

MEDIUM WORKED EXAMPLE - Prove the intersection identity

Problem. Show $P(A \cap B) = P(A) \cap P(B)$.

Solution.

$X \in P(A \cap B)$ iff $X \subseteq A \cap B$ iff $X \subseteq A$ and $X \subseteq B$ iff $X \in P(A) \cap P(B)$.

HARD WORKED EXAMPLE - Disprove a union equality

Problem. Let $A = \{1\}$ and $B = \{2\}$. Compare $P(A \cup B)$ with $P(A) \cup P(B)$.

Solution.

The left side includes $\{1, 2\}$; the right side is $\{\emptyset, \{1\}, \{2\}\}$ and does not. Hence they are unequal.

21. Algebra of set operations

DEFINITION - Algebra of set operations

Set expressions simplify through membership logic and standard identities.

Commutative, associative, distributive, identity, domination, idempotent, complement, absorption and De Morgan laws follow from membership equivalences.

$$A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$$

$$A \cup (A \cap B) = A; \quad A \cap (A \cup B) = A$$

EXAM NOTE - Parentheses matter

Difference is not associative. Preserve the given brackets.

MEDIUM WORKED EXAMPLE - Simplify by laws

Problem. Simplify $(A \cap B) \cup (A \cap B')$.

Solution.

Distribute A: $A \cap (B \cup B') = A \cap U = A$.

HARD WORKED EXAMPLE - Prove a difference identity

Problem. Prove $A - (B \cup C) = (A - B) \cap (A - C)$.

Solution.

For any x , the left means $x \in A$ and x is in neither B nor C , exactly the condition $x \in A - B$ and $x \in A - C$.

22. Integrated finite-set problem solving

DEFINITION - Integrated finite-set problem solving

A complete solution translates words into regions, applies cardinality constraints, and checks every region.

For three sets, begin with the triple intersection, then pair-only regions, single-only regions, then outside.

$$|A \cup B \cup C| = |A| + |B| + |C| - |A \cap B| - |A \cap C| - |B \cap C| + |A \cap B \cap C|$$

EXAM NOTE - Intersections are inclusive

Unless a question says “only”, $|A \cap B|$ includes members also in C.

MEDIUM WORKED EXAMPLE - Two-set survey

Problem. Among 80 users, 46 use A, 39 use B, and 12 use neither. Find both and only A.

Solution.

The union is 68. Thus $\text{both} = 46 + 39 - 68 = 17$, and $\text{only A} = 46 - 17 = 29$.

HARD WORKED EXAMPLE - Three-set reconstruction

Problem. In U of size 100: $|A|=55$, $|B|=48$, $|C|=45$, pairwise intersections are 25, 20, 18, and triple intersection is 10. Find the union and exactly one.

Solution.

Union $= 55 + 48 + 45 - 25 - 20 - 18 + 10 = 95$. Only A $= 20$, only B $= 15$, only C $= 17$; therefore exactly one $= 52$.

Master operation matrix

Use this as a translation table when simplifying expressions or writing element-based proofs.

SET STATEMENT	ELEMENT LOGIC	READING
$x \in A \cup B$	$x \in A$ or $x \in B$	inclusive or
$x \in A \cap B$	$x \in A$ and $x \in B$	both conditions
$x \in A - B$	$x \in A$ and $x \notin B$	left only
$x \in A'$	$x \in U$ and $x \notin A$	outside A, inside U
$x \in A + B$	exactly one of $x \in A$, $x \in B$	exclusive or
$A \subseteq B$	for all x , $x \in A$ implies $x \in B$	containment
$A = B$	$A \subseteq B$ and $B \subseteq A$	mutual containment
$X \in \mathcal{P}(A)$	$X \subseteq A$	power-set bridge

EXAM NOTE - Operation hierarchy

Simplify brackets first. Translate complements and differences carefully. Do not invent arithmetic precedence rules for set operators.

Core identities

$$A \cup \emptyset = A; A \cap U = A; A \cup U = U; A \cap \emptyset = \emptyset$$

$$A \cup A = A; A \cap A = A; A \cup A' = U; A \cap A' = \emptyset$$

$$(A')' = A; (A \cup B)' = A' \cap B'; (A \cap B)' = A' \cup B'$$

$$A - B = A \cap B'; A + B = (A - B) \cup (B - A)$$

$$A \cup (A \cap B) = A; A \cap (A \cup B) = A$$

Exam traps and precision checklist

COMMON TRAP - Element versus subset

If $A = \{1, \{1\}\}$, then $1 \in A$ and $\{1\} \in A$. Also $\{1\} \subseteq A$. Membership and containment are different relation types.

COMMON TRAP - Empty set versus singleton

$\emptyset$ has no elements. $\{\emptyset\}$ has one element. Thus $\emptyset \neq \{\emptyset\}$, although $\emptyset \subset \{\emptyset\}$.

COMMON TRAP - Proper subset

$A \subseteq A$ is true; $A \subset A$ is false. To prove $A \subset B$, supply an element of $B - A$.

COMMON TRAP - Complement needs U

Do not calculate A' until the universal set is known.

COMMON TRAP - Union versus symmetric difference

$A \cup B$ includes the overlap; $A + B$ excludes it.

COMMON TRAP - Difference direction

$A - B$ and $B - A$ answer different questions. Start from the left.

COMMON TRAP - Power-set levels

Elements of $P(A)$ are subsets of A . Use $X \in P(A)$ iff $X \subseteq A$.

Before submitting

- Have I stated the domain or universal set?
- Have I preserved every pair of braces?
- Have I used inclusive “or” for union and exclusive “or” for symmetric difference?
- If claiming equality, have I justified both directions?
- If claiming non-containment, have I supplied a counterexample?
- If counting, have I removed repetitions and simplified equal entries?

Final integrated challenge

Work from definitions. This challenge controls levels of membership, containment, complement and power-set notation at the same time.

DEFINITION - Given

$U = \{\emptyset, \{\emptyset\}, 1, \{1\}, 2\}$, $A = \{\emptyset, 1, \{1\}\}$, and $B = \{\{\emptyset\}, \{1\}, 2\}$.

Questions

- 1. Find $A \cap B$, $A \cup B$, $A - B$, $B - A$ and $A + B$.
- 2. Find A' and B' relative to U .
- 3. Decide whether $\{\emptyset, 1\} \in P(A)$, whether $\{\emptyset, 1\} \subset A$, and whether $\emptyset \in A$.
- 4. Find $|P(A \cap B)|$ and $|P(A \cup B)|$ without listing all subsets.
- 5. Verify $P(A \cap B) = P(A) \cap P(B)$ for these sets.

Fully worked solution

The only common top-level element is $\{1\}$, so $A \cap B = \{\{1\}\}$. The union is U . Thus $A - B = \{\emptyset, 1\}$; $B - A = \{\{\emptyset\}, 2\}$; and $A + B = \{\emptyset, \{\emptyset\}, 1, 2\}$.

$A' = U - A = \{\{\emptyset\}, 2\}$. Similarly $B' = \{\emptyset, 1\}$.

$\{\emptyset, 1\} \subseteq A$, so $\{\emptyset, 1\} \in P(A)$; it misses $\{1\}$, so it is a proper subset of A . Also $\emptyset \in A$ because it is explicitly listed.

$|A \cap B| = 1$, so $|P(A \cap B)| = 2$. The union has five elements, so $|P(A \cup B)| = 2^5 = 32$.

$A \cap B = \{\{1\}\}$, hence $P(A \cap B) = \{\emptyset, \{\{1\}\}\}$. A subset lies in both $P(A)$ and $P(B)$ exactly when all its elements are common to A and B ; only $\emptyset$ and $\{\{1\}\}$ qualify.

EXAM NOTE - Check the levels

The common element is $\{1\}$. Therefore the singleton subset containing that element is $\{\{1\}\}$.

Closed-book exit test

Attempt this without the notes, then use the memorandum.

- 1. Define $A \subseteq B$ and $A \subset B$ precisely.
- 2. Write $\{-5, -3, -1, 1, 3, 5\}$ in set-builder notation.
- 3. Let $A = \{\emptyset, \{\emptyset\}\}$. Decide $\emptyset \in A$, $\{\emptyset\} \subseteq A$ and $\{\{\emptyset\}\} \in P(A)$.
- 4. With $U = \{1, \dots, 10\}$, $A = \{2, 4, 6, 8, 10\}$, find A' .
- 5. Simplify $(A \cup B) \cap (A \cup B')$.
- 6. If $|A| = 25$, $|B| = 19$ and $|A \cup B| = 36$, find $|A \cap B|$.
- 7. List $P(\{a, b, c\})$ and state its cardinality.
- 8. Is $P(A) \cup P(B) = P(A \cup B)$ always true? Justify.
- 9. Prove $A - (B \cap C) = (A - B) \cup (A - C)$.
- 10. Explain why $\{3, 5, 7, \dots\}$ is ambiguous.

Concise memorandum

1. $A \subseteq B$ iff every A-element is a B-element; $A \subset B$ iff $A \subseteq B$ and $A \neq B$.
2. $\{x \in \mathbb{Z} \mid x \text{ is odd and } -5 \leq x \leq 5\}$.
3. True, true, true: $\{\{\emptyset\}\} \subseteq A$, hence $\{\{\emptyset\}\} \in P(A)$.
4. $\{1, 3, 5, 7, 9\}$.
5. $A \cup (B \cap B') = A \cup \emptyset = A$.
6. $25 + 19 - 36 = 8$.
7. $\emptyset, \{a\}, \{b\}, \{c\}, \{a, b\}, \{a, c\}, \{b, c\}, \{a, b, c\}$; cardinality 8.
8. No. A mixed subset can belong only to $P(A \cup B)$.
9. Translate membership and use not-(B and C) iff not-B or not-C.
10. The same opening terms fit different rules, including odd numbers and primes.